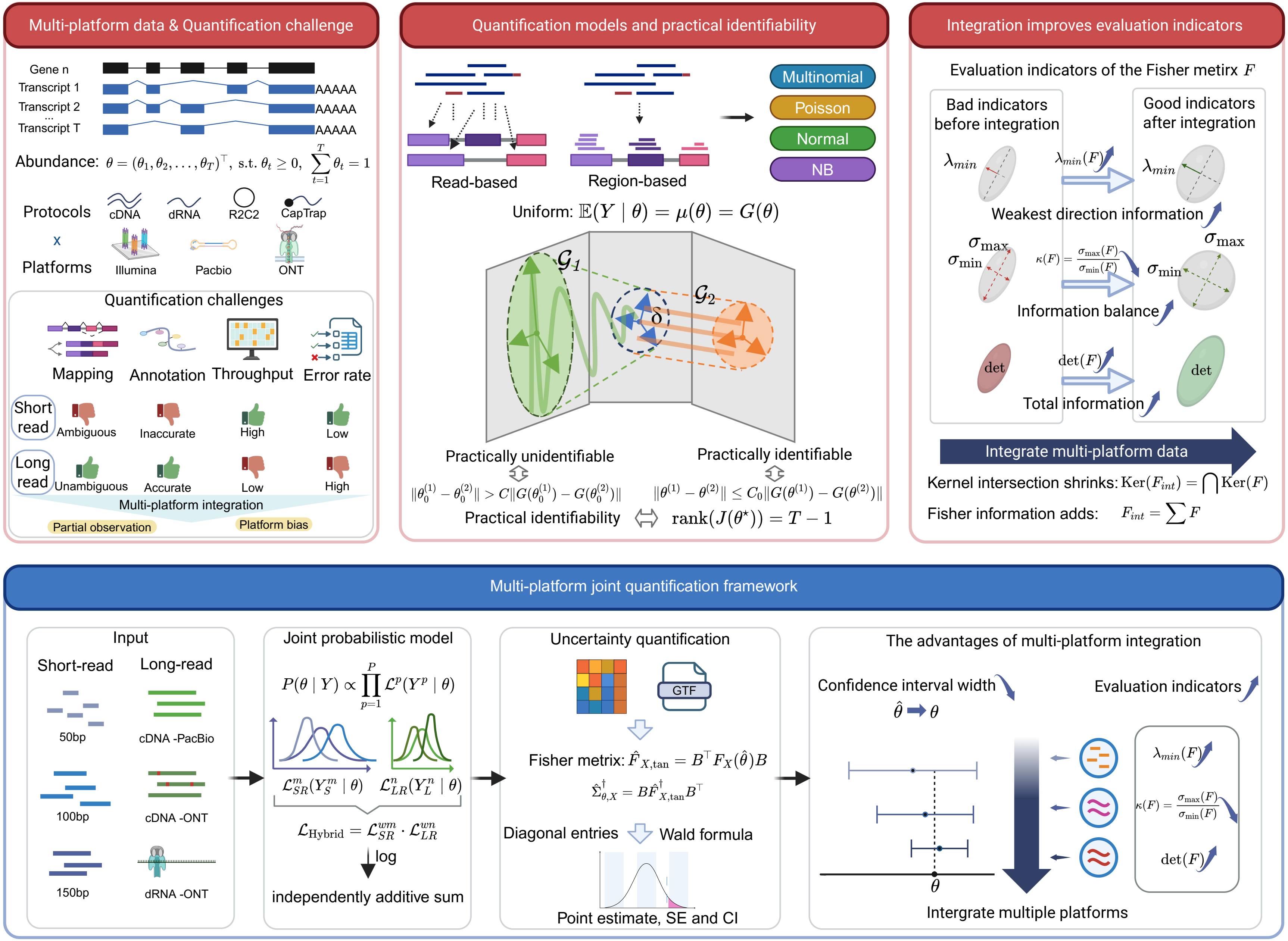
Multi-platform data & Quantification challenge
Gene n
Transcript 1
Transcript 2
Transcript T
AAAAA
Abundance: $\theta = (\theta_1, \theta_2, \ldots, \theta_T)^\top$, s.t. $\theta_t \geq 0$, $\sum_{t=1}^{T} \theta_t = 1$
Protocols
cDNA
dRNA
R2C2
CapTrap
x
Platforms
Illumina
Pacbio
ONT
Quantification challenges
Mapping
Annotation
Throughput
Error rate
Short read
Ambiguous
Inaccurate
High
Low
Long read
Unambiguous
Accurate
Low
High
Multi-platform integration
Partial observation
Platform bias
Quantification models and practical identifiability
Read-based
Region-based
Multinomial
Poisson
Normal
NB
Uniform: $\mathbb{E}(Y \mid \theta) = \mu(\theta) = G(\theta)$
$\mathcal{G}_1$
$\mathcal{G}_2$
δ
Practically unidentifiable
Practically identifiable
$\|\theta_0^{(1)} - \theta_0^{(2)}\| > C\|G(\theta_0^{(1)}) - G(\theta_0^{(2)})\|$
$\|\theta^{(1)} - \theta^{(2)}\| \leq C_0\|G(\theta^{(1)}) - G(\theta^{(2)})\|$
Practical identifiability $\Leftrightarrow$ $\mathrm{rank}(J(\theta^\star)) = T - 1$
Integration improves evaluation indicators
Evaluation indicators of the Fisher metirx F
Bad indicators before integration
Good indicators after integration
λ_{min}
$\lambda_{min}(F)$
Weakest direction information
$\sigma_{\max}$
$\sigma_{\min}$
$\kappa(F) = \frac{\sigma_{\max}(F)}{\sigma_{\min}(F)}$
Information balance
det
$\det(F)$
Total information
Integrate multi-platform data
Kernel intersection shrinks: $\mathrm{Ker}(F_{int}) = \bigcap \mathrm{Ker}(F)$
Fisher information adds: $F_{int} = \sum F$
Multi-platform joint quantification framework
Input
Short-read
Long-read
50bp
100bp
150bp
cDNA -PacBio
cDNA -ONT
dRNA -ONT
Joint probabilistic model
$P(\theta \mid Y) \propto \prod_{p=1}^{P} \mathcal{L}^p(Y^p \mid \theta)$
$\mathcal{L}_{SR}^m(Y_S^m \mid \theta)$
$\mathcal{L}_{LR}^n(Y_L^n \mid \theta)$
$\mathcal{L}_{\text{Hybrid}} = \mathcal{L}_{SR}^{wm} \cdot \mathcal{L}_{LR}^{wn}$
log
independently additive sum
Uncertainty quantification
GTF
Fisher metrix: $\hat{F}_{X,\tan} = B^\top F_X(\hat{\theta})B$
$\hat{\Sigma}_{\theta,X}^\dagger = B\hat{F}_{X,\tan}^\dagger B^\top$
Diagonal entries
Wald formula
Point estimate, SE and CI
The advantages of multi-platform integration
Confidence interval width
$\hat{\theta} \rightarrow \theta$
Evaluation indicators
θ
$\lambda_{min}(F)$
$\kappa(F) = \frac{\sigma_{\max}(F)}{\sigma_{\min}(F)}$
$\det(F)$
Intergrate multiple platforms